NEET UG 2027: Physics Formula Compendium

Module 03: Electrodynamics, Magnetism, EMI, AC & EM Waves

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Strict NCERT & NMC Alignment | Complete Mathematical Variations, Use-Cases & Traps

1 Electrostatics & Capacitance (Unit 11)

1.1 Coulomb's Law, Electric Field & Dipoles

Coulomb's Law, Medium Effect & Dipole Dynamics

$$\mathbf{F}_{12} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}\hat{\mathbf{r}}_{12}, \qquad F_{\text{medium}} = \frac{F_{\text{vacuum}}}{\kappa} = \frac{F_0}{\varepsilon_r} \qquad \left[\frac{1}{4\pi\varepsilon_0} = 8.988\times10^9\,\text{N m}^2\,\text{C}^{-2}\right] \tag{1}$$

$$\mathbf{p} = q(2\mathbf{a}), \qquad \boldsymbol{\tau} = \mathbf{p}\times\mathbf{E}, \qquad U = -\mathbf{p}\cdot\mathbf{E} \qquad [\mathbf{p}\text{ directed from } -q \text{ to } +q] \tag{2}$$

$$E_{\text{axial}} = \frac{1}{4\pi\varepsilon_0}\frac{2pr}{(r^2-a^2)^2} \approx \frac{2kp}{r^3} \qquad [r \gg a;\ \text{direction parallel to } \mathbf{p}] \tag{3}$$

$$E_{\text{equatorial}} = \frac{1}{4\pi\varepsilon_0}\frac{p}{(r^2+a^2)^{3/2}} \approx \frac{kp}{r^3} \qquad [r \gg a;\ \text{direction antiparallel to } \mathbf{p}] \tag{4}$$

$$E(r,\theta) = \frac{kp}{r^3}\sqrt{1+3\cos^2\theta}, \qquad \tan\alpha = \frac{1}{2}\tan\theta \qquad [\theta = \text{angle with dipole axis}] \tag{5}$$

Work Done in Rotating an Electric Dipole from θ_1 to θ_2:

$$W = U_f - U_i = pE(\cos\theta_1 - \cos\theta_2) \implies W_{0\to180^\circ} = 2pE \quad [\text{Max work; unstable equilibrium}] \tag{6}$$

1.2 Gauss's Law & Symmetrical Charge Distributions

Gauss's Theorem & Canonical Field Formulations

$$\Phi_E = \oint \mathbf{E}\cdot d\mathbf{A} = \frac{q_{\text{enclosed}}}{\varepsilon_0} \tag{7}$$

Symmetrical Electric Field Solutions:

$$E_{\text{line}} = \frac{\lambda}{2\pi\varepsilon_0 r} = \frac{2k\lambda}{r} \qquad \text{[Infinitely Long Uniform Line Charge]} \tag{8}$$

$$E_{\text{sheet}} = \frac{\sigma}{2\varepsilon_0} \qquad \text{[Infinite Non-Conducting Plane Sheet (independent} \tag{9}$$

$$E_{\text{conductor}} = \frac{\sigma}{\varepsilon_0} \qquad \text{[Just outside any charged conducting surface]} \tag{10}$$

$$E_{\text{shell}}(r) = \begin{cases} 0 & (r < R) \\ \frac{kq}{R^2} & (r = R) \\ \frac{kq}{r^2} & (r > R) \end{cases}, \qquad V_{\text{shell}}(r) = \begin{cases} \frac{kq}{R} & (r \le R) \\ \frac{kq}{r} & (r > R) \end{cases} \qquad \text{[Thin Spherical Shell of Radius } R\text{]} \tag{11}$$

Uniformly Charged Solid Non-Conducting Sphere (Total Charge q, Radius R):

$$E_{\text{inside}}(r) = \frac{kqr}{R^3} = \frac{\rho r}{3\varepsilon_0}, \qquad V_{\text{inside}}(r) = \frac{kq}{2R^3}(3R^2 - r^2) \implies V_{\text{center}} = \frac{3}{2}\frac{kq}{R} = 1.5\,V_{\text{surface}} \tag{12}$$

1.3 Capacitance, Dielectric Insertion & Stored Energy

Capacitance, Energy Density & Force between Plates

$$C_0 = \frac{\varepsilon_0 A}{d}, \qquad C_{\text{dielectric}} = \frac{\varepsilon_0 A}{d - t\left(1 - \frac{1}{\kappa}\right)}, \qquad C_{\text{conducting slab}} = \frac{\varepsilon_0 A}{d - t} \tag{13}$$

$$U = \frac{1}{2}CV^2 = \frac{Q^2}{2C} = \frac{1}{2}QV, \qquad u_E = \frac{U}{\text{Volume}} = \frac{1}{2}\varepsilon_0 E^2 \quad [\text{J m}^{-3}] \tag{14}$$

$$F_{\text{attraction}} = \frac{Q^2}{2\varepsilon_0 A} = \frac{1}{2}QE = \frac{1}{2}\varepsilon_0 A E^2, \qquad P_{\text{electrostatic}} = \frac{F}{A} = \frac{1}{2}\varepsilon_0 E^2 \tag{15}$$

Common Potential & Heat Dissipation on Connecting Two Capacitors:

$$V_{\text{common}} = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}, \qquad \Delta U_{\text{loss}} = \frac{C_1C_2}{2(C_1 + C_2)}(V_1 - V_2)^2 \tag{16}$$

Capacitor Traps: Force between Plates & Common Potential

- **Factor of $1/2$ in Plate Force:** $F = \frac{1}{2}QE$, NOT QE. One plate experiences the field produced strictly by the **other** plate ($E/2 = \frac{\sigma}{2\varepsilon_0}$), not the total field E.
- **Opposite Polarity Connection:** If positive plate of C_1 is connected to negative plate of C_2, the numerator of common potential becomes $(C_1V_1 - C_2V_2)$ and energy loss is $\frac{C_1C_2}{2(C_1+C_2)}(V_1+V_2)^2$.

Physical Parameter	Case A: Battery REMAINING CONNECTED	Case B: Battery DISCONNECTED
Capacitance (C)	Increases by κ ($C = \kappa C_0$)	Increases by κ ($C = \kappa C_0$)
Potential Difference (V)	Remains Constant ($V = V_0$)	Decreases by κ ($V = V_0/\kappa$)
Electric Field ($E = V/d$)	Remains Constant ($E = E_0$)	Decreases by κ ($E = E_0/\kappa$)
Stored Charge (Q)	Increases by κ ($Q = \kappa Q_0$)	Remains Constant ($Q = Q_0$)
Stored Energy (U)	Increases by κ ($U = \kappa U_0$)	Decreases by κ ($U = U_0/\kappa$)

Table 1: Effect of Inserting a Dielectric Slab (κ) into a Parallel Plate Capacitor.

2 Current Electricity (Unit 12)

2.1 Microscopic Conduction, Drift Velocity & Resistance

Microscopic Drift, Ohm's Law & Temperature Dependence

$$I = neAv_d, \qquad v_d = \frac{eE\tau}{m} = \frac{eV\tau}{mL}, \qquad J = \frac{I}{A} = nev_d = \sigma E = \frac{E}{\rho} \tag{17}$$

$$\mu = \frac{v_d}{E} = \frac{e\tau}{m} \quad [\mathrm{m^2\,V^{-1}\,s^{-1}}], \qquad \rho = \frac{m}{ne^2\tau}, \qquad R = \rho\frac{L}{A} \tag{18}$$

$$R_T = R_0(1+\alpha\Delta T), \qquad \rho_T = \rho_0(1+\alpha\Delta T) \quad [\alpha > 0 \text{ metals; } \alpha < 0 \text{ semiconductors}] \tag{19}$$

Stretching of Wires (Constant Volume $V = A \cdot L = \text{const}$):

$$R \propto L^2 \propto \frac{1}{A^2} \propto \frac{1}{r^4} \implies \text{If } L \to nL, \quad R' = n^2R; \quad \text{For small } \%\Delta L < 5\%, \quad \frac{\Delta R}{R} \approx 2\frac{\Delta L}{L} \tag{20}$$

2.2 Cells, Internal Resistance & Power Transfer

Terminal EMF, Cell Combinations & Max Power Theorem

$$V = \mathcal{E} - Ir \quad [\text{Discharging; } V < \mathcal{E}], \qquad V = \mathcal{E} + Ir \quad [\text{Charging; } V > \mathcal{E}] \tag{21}$$

$$r = R\left(\frac{\mathcal{E}}{V} - 1\right) = R\left(\frac{l_1 - l_2}{l_2}\right) \quad [\text{Internal Resistance Formulat} \tag{22}$$

$$\mathcal{E}_{\text{eq, parallel}} = \frac{\frac{\mathcal{E}_1}{r_1} + \frac{\mathcal{E}_2}{r_2}}{\frac{1}{r_1} + \frac{1}{r_2}} = \frac{\mathcal{E}_1 r_2 + \mathcal{E}_2 r_1}{r_1 + r_2}, \qquad r_{\text{eq, parallel}} = \frac{r_1 r_2}{r_1 + r_2} \tag{23}$$

Maximum Power Transfer Theorem (Load R connected to cell $\mathcal{E}, r$):

$$P_{\text{load}} = I^2 R = \frac{\mathcal{E}^2 R}{(R+r)^2} \implies P = P_{\max} \iff R = r \implies P_{\max} = \frac{\mathcal{E}^2}{4r} \tag{24}$$

2.3 Bridge Circuits & Electrical Instruments

Wheatstone Bridge & Metre Bridge Balance Equations

$$\frac{P}{Q} = \frac{R}{S} \iff I_G = 0 \quad \text{[Wheatstone Balance Condition: Galv. current is null]} \tag{25}$$

Metre Bridge Unknown Resistance S (Balance length l from left):

$$\frac{R}{S} = \frac{l}{100-l} \implies S = R\left(\frac{100-l}{l}\right) \tag{26}$$

Current Electricity Traps: Cell Grouping & Terminal Voltage

- **Reversed Cell in Series Battery:** In a combination of n identical cells each of EMF $\mathcal{E}$ and internal resistance r, if m cells are connected with reversed polarity:

$$\mathcal{E}_{\text{net}} = (n-2m)\mathcal{E}, \qquad r_{\text{net}} = nr \quad \text{(Internal resistances always add!)}$$

- **Short Circuit Terminal Potential:** When cell is short-circuited ($R = 0$), current is maximum $I_{\text{sc}} = \mathcal{E}/r$, and terminal voltage drops to **zero** ($V = \mathcal{E} - I_{\text{sc}} r = 0$).

3 Magnetic Effects of Current & Magnetism (Unit 13)

3.1 Biot-Savart Law & Ampere's Circuital Law

Biot-Savart Law & Symmetrical Field Equations

$$d\mathbf{B} = \frac{\mu_0}{4\pi}\frac{I d\mathbf{l} \times \hat{\mathbf{r}}}{r^2} \qquad \left[\frac{\mu_0}{4\pi} = 10^{-7}\,\mathrm{T\,m\,A^{-1}}\right] \tag{27}$$

$$B_{\text{straight, finite}} = \frac{\mu_0 I}{4\pi d}(\sin\phi_1 + \sin\phi_2) \implies B_{\text{infinite}} = \frac{\mu_0 I}{2\pi d}, \quad B_{\text{semi-infinite}} = \frac{\mu_0 I}{4\pi d} \tag{28}$$

$$B_{\text{circular coil, axis}} = \frac{\mu_0 N I R^2}{2(R^2+x^2)^{3/2}} \implies B_{\text{center}} = \frac{\mu_0 N I}{2R} \quad (x=0) \tag{29}$$

$$B_{\text{arc}} = \frac{\mu_0 I \theta}{4\pi R} \quad [\theta \text{ in radians}], \qquad B_{\text{solenoid}} = \mu_0 n I = \mu_0 \left(\frac{N}{L}\right) I \tag{30}$$

$$\oint \mathbf{B}\cdot d\mathbf{l} = \mu_0 I_{\text{enclosed}} \qquad \text{[Ampere's Circuital Law]} \tag{31}$$

3.2 Lorentz Force, Particle Trajectories & Force on Conductors

Lorentz Equation & Charged Particle Dynamics in Magnetic Fields

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v}\times\mathbf{B}) \implies \mathbf{F}_B = q(\mathbf{v}\times\mathbf{B}) \implies W_B = \int \mathbf{F}_B \cdot d\mathbf{r} = 0 \quad [\text{Speed } v = \text{const}] \tag{32}$$

$$r = \frac{mv}{qB} = \frac{p}{qB} = \frac{\sqrt{2mK}}{qB} = \frac{\sqrt{2mqV}}{qB} \qquad \text{[Circular Orbit Rad} \tag{33}$$

$$T = \frac{2\pi m}{qB}, \qquad \nu = \frac{qB}{2\pi m} \qquad \text{[Period and Frequer} \tag{34}$$

$$p_{\text{pitch}} = v_\parallel T = \frac{2\pi m v\cos\theta}{qB} \qquad \text{[Helical Path Pitch} \tag{35}$$

$$\mathbf{F} = I(\mathbf{L}\times\mathbf{B}), \qquad \frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d} \qquad \text{[Parallel attract; an} \tag{36}$$

3.3 Galvanometer Conversions & Magnetic Materials

Moving Coil Galvanometer, Ammeter, Voltmeter & Magnetic Matter

$$\tau = NIAB = C\theta \implies I = \left(\frac{C}{NAB}\right)\theta, \quad I_s = \frac{\theta}{I} = \frac{NAB}{C}, \quad V_s = \frac{\theta}{V} = \frac{I_s}{R_G} \tag{37}$$

$$S = \frac{I_g R_g}{I - I_g} \quad \text{[Ammeter Shunt in Parallel]}, \qquad R_s = \frac{V}{I_g} - R_g \quad \text{[Voltmeter Multiplier in Series]} \tag{38}$$

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) = \mu H, \qquad \chi = \frac{M}{H}, \qquad \mu_r = 1 + \chi = \frac{\mu}{\mu_0} \tag{39}$$

Curie's Law for Paramagnetic Materials & Curie-Weiss Law for Ferromagnets:

$$\chi_{\text{para}} = \frac{C}{T} \implies \chi \propto \frac{1}{T}, \qquad \chi_{\text{ferro}} = \frac{C'}{T - T_c} \quad (T > T_c) \tag{40}$$

Magnetism Traps: Work Done by Magnetic Force & Galvanometer Sensitivity

- **Work Done by Magnetic Force is Zero:** Because $\mathbf{F}_B \perp \mathbf{v}$ at every instant, $\mathbf{F}_B \cdot \mathbf{v} = 0$. **A static magnetic field does zero work on a charged particle**. It changes the direction of motion, never the kinetic energy or speed.
- **Current Sensitivity vs Voltage Sensitivity:** Doubling the number of turns N doubles current sensitivity ($I_s \propto N$), but resistance also doubles ($R \propto N$), leaving voltage sensitivity $V_s = I_s/R_G$ **completely unchanged**!

4 Electromagnetic Induction & Alternating Current (Unit 14)

4.1 Faraday's Laws, Motional EMF & Inductance

Faraday-Lenz Law, Motional EMF & Self/Mutual Inductance

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -N\frac{d}{dt}(BA\cos\theta), \qquad q_{\text{induced}} = \frac{\Delta\Phi_B}{R} \quad \text{[Independent of time!]} \tag{41}$$

$$\mathcal{E}_{\text{motional}} = Bvl \quad \text{[Translational Rod]}, \qquad \mathcal{E}_{\text{rotational}} = \frac{1}{2}B\omega L^2 \quad \text{[Rotating Rod about one end]} \tag{42}$$

$$\mathcal{E} = -L\frac{dI}{dt}, \qquad U = \frac{1}{2}LI^2, \qquad L_{\text{solenoid}} = \mu_0 n^2 Al = \frac{\mu_0 N^2 A}{l} \tag{43}$$

$$\mathcal{E}_2 = -M\frac{dI_1}{dt}, \qquad M = k\sqrt{L_1 L_2} \quad (0 \le k \le 1) \tag{44}$$

4.2 Alternating Current Circuits, Impedance & Resonance

AC Values, Reactance, Series LCR Circuit & Resonance

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}} \approx 0.707 I_0, \qquad V_{\text{rms}} = \frac{V_0}{\sqrt{2}} \approx 0.707 V_0, \qquad I_{\text{avg, half}} = \frac{2I_0}{\pi} \approx 0.637 I_0 \tag{45}$$

$$X_L = \omega L = 2\pi\nu L, \qquad X_C = \frac{1}{\omega C} = \frac{1}{2\pi\nu C} \quad [X_L \text{ leads by } \pi/2; X_C \text{ lags by } \pi/2] \tag{46}$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}, \qquad \tan\phi = \frac{X_L - X_C}{R}, \qquad V_0 = \sqrt{V_R^2 + (V_L - V_C)^2} \tag{47}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \iff \nu_0 = \frac{1}{2\pi\sqrt{LC}} \quad \text{[Series Resonance Condition: } X_L = X_C\text{]} \tag{48}$$

$$Z_{\min} = R, \quad I_{\max} = \frac{V_{\text{rms}}}{R}, \quad \phi = 0, \quad Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{\nu_0}{\Delta\nu} \tag{49}$$

Average Power, Power Factor & Ideal Transformer:

$$P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos\phi = I_{\text{rms}}^2 R, \qquad \cos\phi = \frac{R}{Z} \quad \text{[Wattless Current } I_w = I_{\text{rms}} \sin\phi\text{]} \tag{50}$$

$$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s} = K, \qquad \eta = \frac{P_{\text{out}}}{P_{\text{in}}} = \frac{V_s I_s}{V_p I_p} \tag{51}$$

5 Electromagnetic Waves (Unit 15)

Displacement Current, Wave Speed, Energy & Radiation Pressure

$$I_d = \varepsilon_0 \frac{d\Phi_E}{dt}, \qquad \oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 \left(I_c + \varepsilon_0 \frac{d\Phi_E}{dt} \right) \quad \text{[Ampere-Maxwell Law]} \tag{52}$$

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = \frac{E_0}{B_0} \approx 3 \times 10^8 \, \mathrm{m\,s^{-1}}, \qquad v_{\text{medium}} = \frac{1}{\sqrt{\mu \varepsilon}} = \frac{c}{\sqrt{\mu_r \varepsilon_r}} \tag{53}$$

$$u_{\text{total}} = u_E + u_B = \frac{1}{2}\varepsilon_0 E_{\text{rms}}^2 + \frac{B_{\text{rms}}^2}{2\mu_0} = \varepsilon_0 E_{\text{rms}}^2 = \frac{B_{\text{rms}}^2}{\mu_0} \quad \text{[Equal Electric \& Magnetic Energy]} \tag{54}$$

$$I = \langle S \rangle = \frac{1}{2}\varepsilon_0 c E_0^2 = \varepsilon_0 c E_{\text{rms}}^2, \qquad P_{\text{rad}} = \begin{cases} \frac{I}{c} & \text{(Total Absorption)} \\ \frac{2I}{c} & \text{(Total Reflection)} \end{cases} \tag{55}$$

Wave Type	**Wavelength (λ)**	**Frequency (ν)**	**Primary Source & High-Yield Application**
γ-Rays	$< 10^{-3}$ nm	$> 10^{20}$ Hz	Radioactive decay; cancer radiotherapy
X-Rays	1 nm – 10^{-3} nm	$10^{18} - 10^{20}$ Hz	Fast electrons hitting heavy metal target; bone imaging
UV Rays	400 nm – 1 nm	$10^{15} - 10^{17}$ Hz	Very hot bodies (Sun), arcs; LASIK eye surgery, germicidal
Visible	700 nm – 400 nm	$4 \times 10^{14} - 7.5 \times 10^{14}$ Hz	Atomic electron transitions; human optical vision
Infrared	1 mm – 700 nm	$10^{11} - 4 \times 10^{14}$ Hz	Hot bodies, molecules; thermal therapy, remote controls
Microwaves	0.1 m – 1 mm	$10^9 - 10^{11}$ Hz	Klystrons, magnetrons; RADAR systems, microwave ovens
Radio Waves	> 0.1 m	$< 10^9$ Hz	Accelerated charges in antennas; cellular/radio broadcasting

Table 2: Electromagnetic Spectrum in Decreasing Order of Frequency / Increasing Wavelength.

End of Module 03 Formula Handbook

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